An Agentic Active Learning-Enhanced Adversarial Defense Framework for Cyber Threat Intelligence Classification

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Abstract

The increasing deployment of artificial intelligence systems in cybersecurity has amplified vulnerability to adversarial attacks that manipulate model behavior through imperceptible input perturbations. While adversarial training improves robustness, existing approaches treat all adversarial samples equally, resulting in inefficient retraining and high annotation costs—particularly problematic in security domains requiring expert labeling. This paper introduces an **Agentic Active Learning-Enhanced Adaptive Defense Framework** that intelligently selects the most informative adversarial examples for retraining using uncertainty sampling. The framework is architected as a multi-agent system of four specialized agents, each with defined roles, perceptions, and goal-directed actions: (1) a *Detection Agent* that perceives incoming inputs and flags adversarial anomalies, (2) a *Selection Agent* that reasons over prediction uncertainty to allocate the labeling budget, (3) a *Retraining Agent* that incrementally adapts model parameters, and (4) an *Audit Agent* that monitors system behavior and generates explainable decision logs. Experiments on the AnnoCTR cyber threat intelligence dataset demonstrate that uncertainty-based selection achieves equivalent robustness to random sampling using only 40% of labeled samples, reducing annotation burden by 60% while decreasing the attack success rate by 66%. The proposed framework bridges the gap between theoretical robustness research and operational cybersecurity requirements, providing a scalable and interpretable paradigm for trustworthy AI defense.

Keywords: Adversarial Machine Learning, Active Learning, Agentic AI, Multi-Agent Systems, Cyber Threat Intelligence, Robustness, Trustworthy AI, Deep Learning, Natural Language Processing, MITRE ATT&CK

1. Introduction

Artificial intelligence has become integral to modern cybersecurity operations, powering systems for intrusion detection, malware analysis, phishing detection, and cyber threat intelligence (CTI) classification. These AI-driven systems analyze vast quantities of security data to identify threats, categorize attack patterns, and support security analysts in making informed decisions. As organizations increasingly rely on machine learning models for security-critical tasks, the reliability and robustness of these systems become paramount concerns.

However, the same machine learning models that enable sophisticated threat detection are themselves vulnerable to adversarial attacks—deliberate, often imperceptible manipulations designed to mislead model predictions [1, 2]. In cybersecurity contexts, adversaries can craft malicious inputs that evade detection systems, manipulate threat classifications, or extract sensitive information about model behavior. Recent studies demonstrate that even state-of-the-art deep learning models for network intrusion detection and natural language processing-based threat analysis exhibit significant vulnerabilities to such attacks [4, 5].

The challenge of adversarial robustness is particularly acute in CTI systems that rely on text classification to map threat reports to standardized frameworks such as MITRE ATT&CK. Threat actors can deliberately obfuscate their techniques through paraphrasing, synonym substitution, or subtle linguistic modifications that preserve malicious intent

while evading automated detection [6]. Unlike traditional image-based adversarial attacks, textual perturbations in security contexts often have semantic grounding—attackers genuinely attempt to disguise their activities through careful word choice and description.

While adversarial training has emerged as a primary defense mechanism, existing approaches face significant practical limitations. Standard adversarial training treats all adversarial examples equally, requiring extensive retraining on large volumes of perturbed samples [3]. In cybersecurity domains, this presents a critical bottleneck: expert annotation of threat intelligence requires specialized domain knowledge, making large-scale labeling prohibitively expensive. Security analysts' time is limited and valuable, yet current defenses demand substantial human effort without optimizing which samples provide the greatest learning value.

Despite significant advances in both adversarial robustness and active learning independently, their integration remains underexplored. Existing literature on adversarial defense focuses primarily on robustness metrics and attack mitigation strategies [8, 9], while active learning research has demonstrated success in reducing annotation costs across various domains [16,17]. However, no prior work has systematically integrated uncertainty-based sample selection into adversarial retraining for cybersecurity AI systems.

This gap is particularly critical in CTI classification, where: (1) expert annotation requires specialized security knowledge, (2) adversarial obfuscation is intentional and evolving, and (3) the cost of misclassification—missing a genuine threat or misattributing an attack technique—carries significant operational consequences. A defense framework that maximizes robustness improvement per labeled sample would directly address these practical constraints.

This paper makes the following contributions:

1. We propose an **Agentic Multi-Agent Defense Framework** that formalizes four specialized agents—Detection, Selection, Retraining, and Audit—each with explicit roles, perceptions, actions, and goals. This agentic architecture integrates active learning for intelligent adversarial sample selection, enabling efficient retraining with reduced annotation overhead.
2. We introduce an **uncertainty-based sampling strategy** specifically designed for adversarial retraining in NLP-based cyber threat detection systems, using prediction entropy to identify the most informative samples.
3. We conduct **extensive experiments** on the AnnoCTR dataset, demonstrating 60% reduction in labeling cost while achieving comparable or superior robust accuracy to random sampling approaches.
4. We provide a **comprehensive analysis** of the robustness-efficiency trade-off, offering practical guidelines for real-world deployment in security operations centers.

The remainder of this paper is organized as follows. Section 2 provides background on adversarial attacks, defense mechanisms, and active learning fundamentals. Section 3 reviews related work from 2022–2025 and identifies research gaps. Section 4 presents the proposed framework architecture in detail. Section 5 describes the experimental setup. Section 6 reports and analyzes results. Section 7 discusses implications and limitations. Section 8 concludes with future research directions.

2. Background and Preliminaries

This section establishes the theoretical foundations for the proposed framework, covering adversarial attacks in AI systems, defense mechanisms, active learning fundamentals, and the cyber threat intelligence domain.

2.1 Adversarial Attacks in AI Systems

Adversarial examples are inputs deliberately crafted to cause machine learning models to produce incorrect outputs while appearing normal to human observers [1]. Formally, given a classifier f_θ parameterized by θ, a clean input $\mathbf{x}$ with true label y, an adversarial example $\mathbf{x}_{adv}$ is defined as:

$$\mathbf{x}_{adv} = \mathbf{x} + \boldsymbol{\delta}, \quad \text{s.t.} \quad \|\boldsymbol{\delta}\|_p \leq \epsilon \tag{1}$$

where $\boldsymbol{\delta}$ is the adversarial perturbation bounded by ϵ under the L_p norm (typically L_∞ or L_2), and the attack succeeds when $f_\theta(\mathbf{x}_{adv}) \neq y$.

Gradient-Based Attacks: The Fast Gradient Sign Method (FGSM) [1] generates perturbations aligned with the gradient of the loss function:

$$\boldsymbol{\delta} = \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\theta, \mathbf{x}, y)) \tag{2}$$

Projected Gradient Descent (PGD) [3] extends FGSM through iterative optimization:

$$\mathbf{x}^{(t+1)} = \Pi_{\mathbf{x}+\mathcal{S}} \left(\mathbf{x}^{(t)} + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\theta, \mathbf{x}^{(t)}, y)) \right) \tag{3}$$

where Π denotes projection onto the ϵ-ball $\mathcal{S}$ around the original input.

Text-Based Attacks: In NLP contexts, adversarial perturbations take forms including synonym substitution, character-level modifications, paraphrasing, and semantic-preserving transformations [6]. Unlike continuous image perturbations, textual attacks operate in discrete token space, requiring specialized generation techniques.

Attack Taxonomy: Adversarial attacks are categorized by adversary knowledge (white-box vs. black-box), attack timing (training-time poisoning vs. inference-time evasion), and attack goal (targeted vs. untargeted) [7]. In cybersecurity contexts, evasion attacks at inference time pose the most immediate operational threat.

2.2 Defense Mechanisms Overview

Adversarial Training: The dominant defense paradigm formulates robustness as a min-max optimization problem [3]:

$$\min_{\theta} \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}} \left[\max_{\|\boldsymbol{\delta}\|\leq\epsilon} \mathcal{L}(\theta, \mathbf{x} + \boldsymbol{\delta}, y) \right] \tag{4}$$

This approach trains the model on worst-case perturbations, improving robustness but at significant computational cost and potential degradation of clean accuracy.

Input Preprocessing: Techniques such as input transformation, feature squeezing, and denoising aim to remove adversarial perturbations before classification [13]. While computationally efficient, these methods provide limited defense against adaptive attacks.

Detection-Based Defenses: Rather than achieving robustness through training, detection approaches identify adversarial inputs at inference time using statistical analysis, feature inconsistency, or auxiliary classifiers [5]. Explainable AI (XAI) methods increasingly support detection by highlighting suspicious activation patterns.

2.3 Active Learning Fundamentals

Active learning is a machine learning paradigm where the algorithm selectively queries an oracle (typically a human annotator) for labels on the most informative unlabeled samples [16]. This approach is particularly valuable when labeling is expensive or time-consuming.

Pool-Based Active Learning: Given a small labeled set $\mathcal{L}$, a large unlabeled pool $\mathcal{U}$, and a labeling budget B, the goal is to select samples $\mathcal{S}^* \subset \mathcal{U}$ where $|\mathcal{S}^*| = B$ that maximize model improvement.

Uncertainty Sampling: The most common query strategy selects samples where the model is least confident. Using prediction entropy as the uncertainty measure:

$$\mathbf{x}^* = \arg\max_{\mathbf{x}\in\mathcal{U}} H(P(y|\mathbf{x})) = \arg\max_{\mathbf{x}\in\mathcal{U}} \left[-\sum_{c=1}^{C} P(y=c|\mathbf{x}) \log P(y= \tag{5}$$

where C is the number of classes and $P(y = c|\mathbf{x})$ is the model's predicted probability for class c.

Margin Sampling: An alternative strategy selects samples with the smallest margin between the top two predicted classes:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{U}} [P(y_1|\mathbf{x}) - P(y_2|\mathbf{x})] \tag{6}$$

where y_1 and y_2 are the first and second most probable classes.

Relevance to Adversarial Defense: Active learning naturally applies to adversarial retraining because: (1) adversarial pools contain samples of varying difficulty and informativeness, (2) high-entropy samples often represent novel or boundary-case attacks, and (3) learning from the most confusing examples provides greater robustness improvement than random selection.

2.4 Agentic AI and Multi-Agent Systems

Agentic AI refers to systems composed of autonomous agents that perceive their environment, reason over observations, and execute goal-directed actions [16]. Formally, an agent $\mathcal{A}$ is characterized by a four-tuple $(\mathcal{P}, \mathcal{A}ct, \mathcal{G}, \pi)$, where $\mathcal{P}$ denotes the agent's perceptual inputs, $\mathcal{A}ct$ its action space, $\mathcal{G}$ its operational goal, and π its decision policy mapping perceptions to actions.

In *multi-agent systems* (MAS), multiple agents with complementary specializations collaborate

within a shared environment, communicating through structured interfaces. This paradigm offers several advantages over monolithic architectures: modularity enables independent development and testing of each agent; specialization allows each agent to optimize for a narrow, well-defined sub-task; and the closed-loop interaction between agents supports emergent adaptive behavior [21].

In the context of adversarial defense, the agentic paradigm is particularly well-suited because defense is inherently a multi-phase process—detection, prioritization, adaptation, and oversight—each requiring distinct capabilities. Casting these phases as goal-directed agents makes the division of responsibility explicit, supports human-in-the-loop integration at any stage, and provides a principled foundation for future extensions such as agent-level reinforcement learning or continual adaptation.

2.5 Cyber Threat Intelligence and MITRE ATT&CK

, analyzing, and disseminating information about cyber threats to support defensive decision-making. Modern CTI systems increasingly employ NLP techniques to automatically extract indicators of compromise (IOCs), identify threat actor behaviors, and map reports to standardized frameworks.

The MITRE ATT&CK framework provides a comprehensive knowledge base of adversary tactics, techniques, and procedures (TTPs) observed in real-world attacks. CTI classification systems map threat reports to ATT&CK categories, enabling standardized analysis and threat correlation. However, this classification task is vulnerable to adversarial manipulation, as threat actors may deliberately obfuscate their techniques to evade automated detection.

The AnnoCTR dataset [19] provides labeled cyber threat reports aligned with MITRE ATT&CK categories, enabling research on automated CTI classification and its robustness to adversarial perturbations.

3. Related Work

This section reviews recent advances in adversarial robustness (2022–2025), active learning in security applications, and identifies the research gap addressed by this work.

3.1 Adversarial Robustness in Machine Learning

Early modern efforts prioritized making adversarial robustness practical and provable in unimodal machine learning models. Awais and Bae [**?**] survey computationally efficient robustness techniques, organizing prior work into (i) modifications to adversarial training and (ii) transfer learning strategies that reduce computational cost while preserving robustness. Their analysis highlights that efficiency gains often rely on careful attack sampling and curriculum design; however, robustness may degrade under distribution shifts or changes in adversarial budgets.

Complementary to efficiency-focused studies, Khamaiseh et al. [**?**] present a comprehensive taxonomy of adversarial attacks and defenses, demonstrating how small, loss-aligned perturbations exploit local decision boundary geometry. While foundational, this body of work remains largely vision-centric and focused on controlled, unimodal settings.

Formal verification has emerged as a parallel pillar of robustness research. Meng et al. [**?**] systematize neural network verification through property specification, problem reduction via SMT and MILP encodings, and reasoning strategies such as abstract interpretation. These methods provide provable guarantees under bounded perturbations but face scalability limitations on modern architectures and often rely on conservative assumptions that under-approximate real-world threat models. Collectively, these studies establish efficiency and verifiability as core robustness objectives, primarily in unimodal learning environments.

3.2 Robustness in Cybersecurity Applications

A second research stream integrates adversarial robustness with cybersecurity requirements and trustworthy AI principles, including transparency, accountability, and standardized evaluation. Vassilev et al. [**?**], through the NIST Adversarial Machine Learning taxonomy, codify attacks, mitigations, and threat models, enabling cross-study comparability across security domains. However, this effort stops short of defining certified robustness metrics, particularly for complex or multimodal systems.

Lee et al. [**?**] propose the SARS robustness score, which quantifies robustness by incorporating per-

turbation difficulty and feature-space impact, addressing limitations of accuracy- centric evaluation. While promising, its adoption beyond specific malware detection settings remains limited. Security-driven system designs further illustrate the co-existence of robustness and interpretability. Nguyen et al. [?] introduce an intrusion detection pipeline that couples adversarial training with explainable AI, demonstrating improved resilience to adversarial network traffic while providing human-interpretable rationales. Similarly, Prasad et al. [?] survey NLP-based threat detection methods leveraging meta-learning, reinforcement learning, and self- supervision, emphasizing explainability-driven anomaly identification. Despite these advances, continual retraining costs and domain portability remain open challenges.

3.3 Multimodal and LLM Robustness

As AI systems increasingly integrate text, vision, audio, and other modalities, robustness challenges extend beyond unimodal perturbations to cross-modal and alignment-based attacks. Kapoor et al. [?] provide a practitioner-oriented survey categorizing multimodal attacks across cross-input, fusion, and alignment layers, offering practical guidance for engineering robust multimodal models. Empirical studies highlight concrete vulnerabilities: medical AI surveys report how perturbations in one modality can propagate through diagnostic or report-generation pipelines, while Cui et al. demonstrate image-driven adversarial failures in multimodal large language models.

On the defense side, emerging approaches target multimodal- specific robustness guarantees. Wang et al. [?] propose MMCert, the first certified defense framework designed for paired-modality models, deriving robustness guarantees under bounded perturbations. Chen et al. [?] explore spiking- transformer architectures for multimodal emotion recognition, reporting improved robustness and energy efficiency under FGSM, BIM, and PGD attacks. Jiang et al. [?] survey multimodal large language model robustness across datasets, modalities, and evaluation metrics, emphasizing fragile modality alignment and the lack of standardized benchmarks. Together, these works signal a transition from unimodal optimization toward alignment-aware robustness, though certified and generalizable defenses remain nascent.

3.4 Active Learning for Security and Adversarial Defense

Active learning addresses the growing cost of data annotation and retraining by selectively querying the most informative samples for labeling. Classical strategies such as uncertainty sampling, margin-based selection, and query-by-committee prioritize samples near decision boundaries, where model uncertainty is highest. While well established in general machine learning, active learning has gained relevance in security contexts characterized by evolving threats and scarce expert annotations.

Recent surveys document a shift from classical pool-based active learning toward deep active learning methods designed to scale to high-dimensional and deployment- oriented settings [21].

In cybersecurity applications, active learning has been applied to malware detection, intrusion de- tection, and anomaly analysis to reduce analyst workload while maintaining detection performance. Prior studies show that selective sampling can sub- stantially reduce labeling requirements under non- adversarial or weakly adaptive conditions. However, these approaches rarely consider adversarial settings in which attackers may actively manipulate uncertainty estimates to evade detection.

Despite overlapping goals of efficiency and adaptivity, adversarial robustness and active learning remain largely decoupled in existing literature. Most adversarial training pipelines retrain uniformly on detected adversarial samples, ignoring their relative informativeness. This leads to inefficient retraining cycles, particularly in large-scale or multimodal cybersecurity systems, and highlights a gap at the intersection of robustness, cost efficiency, and adaptive learning.

3.5 Summary and Research Gap

Across 2022–2025, adversarial robustness research has progressed from efficiency- and verification-focused unimodal methods to trustworthy, security-aligned pipelines and, more recently, to multimodal and large language model robustness. Despite these advances, three persistent limitations emerge from the literature. First, adversarial training and formal verification incur high computational and human costs, limiting scalability to modern, large-scale, and multimodal architectures. Second, many robustness evaluations assume static or narrowly de-

fined threat models, reducing generalization to operational cybersecurity environments characterized by evolving adversarial behavior. Third, although active learning provides a principled mechanism for reducing labeling effort and improving adaptivity, it remains largely decoupled from adversarial robustness pipelines in existing work.

Table 1 summarizes representative prior studies and highlights that existing approaches typically address adversarial training, explainability, or active learning in isolation. In particular, active learning techniques are rarely integrated into adversarial retraining loops, resulting in inefficient use of expert annotations during continual defense updates. This fragmentation motivates the need for a unified framework that jointly incorporates adversarial robustness, uncertainty-driven sample selection, and explainable auditing within a single adaptive cybersecurity-oriented AI system.

4. Proposed Multi-Agent Defense Framework

This section presents the **Agentic Active Learning-Enhanced Adaptive Defense Framework**, designed to maximize adversarial robustness while minimizing annotation costs. The framework is architected as a multi-agent system in which four specialized agents—each with explicit perceptions, actions, and goals—collaborate in a continuous closed-loop to detect, prioritize, retrain against, and audit adversarial threats.

4.1 Agent Formalization

Following the standard agent model, each component is formally defined as a tuple $\mathcal{A} = (\mathcal{P}, \mathcal{A}ct, \mathcal{G}, \pi)$. Table 2 summarizes the four agents and their formal properties.

The agents interact through a directed message-passing protocol: the Detection Agent feeds its output pool $\mathcal{U}$ to the Selection Agent; the Selection Agent forwards the prioritized set $\mathcal{S}^*$ to the Retraining Agent; and the Audit Agent receives asynchronous signals from all three operational agents, maintaining a running audit log. A human security analyst serves as the labeling oracle between the Selection and Retraining agents, constituting the human-in-the-loop component of the system.

4.2 Architecture Overview

The multi-agent framework consists of four specialized agents operating in a closed feedback loop, as illustrated in Figure 1:

1. **Detection Agent:** Perceives incoming inputs, performs anomaly analysis, and populates the adversarial candidate pool $\mathcal{U}$.
2. **Selection Agent:** Scores candidates by prediction uncertainty and selects the most informative samples $\mathcal{S}^*$ within the labeling budget B.
3. **Retraining Agent:** Receives labeled samples from the oracle and performs incremental adversarial training to improve model robustness.
4. **Audit Agent:** Asynchronously monitors all agent decisions, tracks robustness metrics, and provides explainable logs for compliance and human oversight.

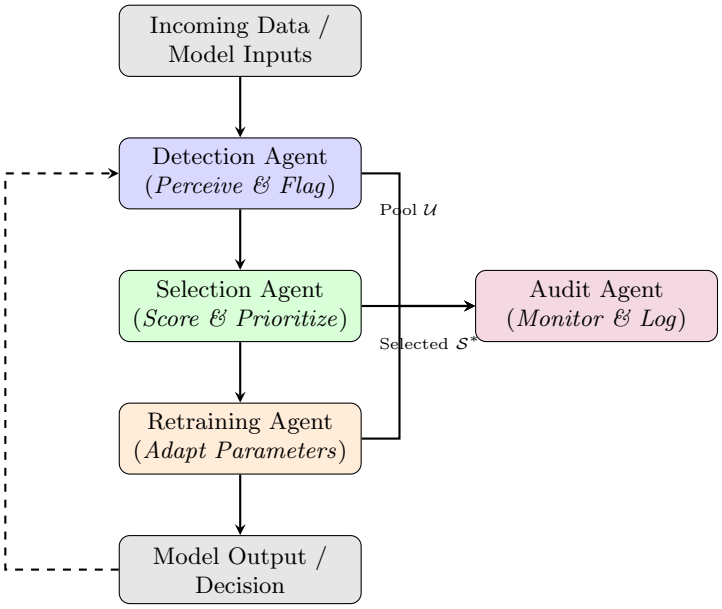

Figure 1: Multi-agent architecture of the Agentic Active Learning-Enhanced Adaptive Defense Framework. Four specialized agents—Detection, Selection, Retraining, and Audit—operate in a closed feedback loop. The Audit Agent receives asynchronous signals from all operational agents. Dashed arrow indicates the continuous feedback loop.

The closed-loop design ensures that the system continuously improves by learning from detected adversarial patterns, with the audit engine providing

Table 1: Comparison with Existing Work and Identified Research Gaps

Work	**Domain**	**Adv. Train**	**Active Learn**	**Integrated Retrain**	**XAI**	**Agentic**
Nguyen *et al.* [5]	IDS	✓	–	–	✓	–
Prasad *et al.* [6]	NLP Sec.	✓	–	–	✓	–
Muhammad & Bae [8]	General	✓	–	–	–	–
Settles [16]	General	–	✓	–	–	–
Gal *et al.* [17]	Vision	–	✓	–	–	–
Ours	**CTI**	✓	✓	✓	✓	✓

Table 2: Multi-Agent Framework: Agent Roles, Perceptions, Actions, and Goals

Agent	**Perception** $\mathcal{P}$	**Action** $\mathcal{A}ct$	**Goal** $\mathcal{G}$
Detection Agent	Raw text inputs; model activations; feature distributions	Flag anomalies; populate candidate pool $\mathcal{U}$	Maximize adversarial recall with low false-positive rate
Selection Agent	Candidate pool $\mathcal{U}$; model confidence scores $P(y\|\mathbf{x})$	Compute entropy $H(\mathbf{x})$; select top-B samples $\mathcal{S}^*$	Maximize robustness gain per labeled sample
Retraining Agent	Labeled set $\mathcal{S}^*$; clean training data; current parameters θ	Incremental fine-tuning; gradient regularization	Improve robust accuracy while preserving clean accuracy
Audit Agent	All agent decisions; performance metrics; model states	Log events; generate explanations; track robustness drift	Ensure transparency, compliance, and human oversight

transparency and enabling human oversight of the adaptation process.

4.3 Detection Agent

Role: *Sentinel.* ***Perception:*** *Raw text inputs, model activation patterns, and feature distributions.* ***Action:*** *Flag anomalous inputs and populate the candidate pool* $\mathcal{U}$*.* ***Goal:*** *Maximize adversarial recall while maintaining a low false-positive rate.*

The Detection Agent serves as the perceptual gateway of the multi-agent system, continuously monitoring incoming inputs to identify potential adversarial samples. It performs three key functions:

Feature Consistency Analysis: The module analyzes input features for statistical deviations from the training distribution. Inputs with unusual feature patterns or activation signatures are flagged as potential adversarial candidates.

XAI-Driven Anomaly Detection: Explainable AI techniques identify inputs where model attention or feature attribution patterns deviate from expected behavior. Unusual explanation patterns may indicate adversarial manipulation even when predictions appear confident.

Candidate Pool Population: Detected anomalies are added to the adversarial candidate pool $\mathcal{U}$, which the Detection Agent forwards to the Selection Agent. The pool accumulates potential adversarial examples for subsequent prioritization and labeling.

4.4 Selection Agent

Role: *Strategist.* ***Perception:*** *Candidate pool* $\mathcal{U}$ *from the Detection Agent; model confidence scores* $P(y|\mathbf{x})$*.* ***Action:*** *Compute prediction entropy* $H(\mathbf{x})$ *for each candidate; select top-B samples* $\mathcal{S}^*$*.* ***Goal:*** *Maximize robustness gain per labeled sample within the annotation budget.*

The Selection Agent represents the core methodological contribution of this work. Rather than treating all adversarial candidates equally, this agent intelligently prioritizes samples that provide maximum learning value for robustness improvement.

4.4.1 Problem Formulation

Given a trained model M_θ, an adversarial candidate pool $\mathcal{U} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ from the detection module, and a labeling budget $B \ll |\mathcal{U}|$, the goal is to select a subset $\mathcal{S}^* \subset \mathcal{U}$ where $|\mathcal{S}^*| = B$ that maximizes robustness improvement per labeled sample.

4.4.2 Uncertainty Scoring Mechanism

For each adversarial candidate $\mathbf{x} \in \mathcal{U}$, we compute uncertainty using prediction entropy:

$$H(\mathbf{x}) = -\sum_{c=1}^{C} P(y = c|\mathbf{x}) \cdot \log P(y = c|\mathbf{x}) \tag{7}$$

where C is the number of classes (MITRE ATT&CK categories) and $P(y = c|\mathbf{x})$ is the softmax probability for class c.

The entropy score has the following interpretation:

- $H(\mathbf{x}) \approx 0$: Model is highly confident (easy sample, low informativeness)
- $H(\mathbf{x}) \approx \log(C)$: Maximum uncertainty (hard sample, high informativeness)

High-entropy samples represent inputs where the model is most confused, typically corresponding to novel attack patterns or samples near decision boundaries—precisely the examples that provide the greatest learning value.

4.4.3 Budget-Constrained Selection

Given the scored pool $\{(\mathbf{x}_i, H(\mathbf{x}_i))\}_{i=1}^n$ and budget B, the selection objective is:

$$\mathcal{S}^* = \arg\max_{\mathcal{S} \subset \mathcal{U}, |\mathcal{S}| = B} \sum_{\mathbf{x} \in \mathcal{S}} H(\mathbf{x}) \tag{8}$$

This reduces to selecting the top-B samples ranked by entropy score.

4.4.4 Algorithm Description

Algorithm 1 presents the complete active adversarial retraining procedure. The algorithm iteratively scores candidates, selects high-uncertainty samples, obtains expert labels, and retrains the model on the selected examples.

Algorithm 1 Active Learning for Adversarial Retraining

Require: Model M_θ, Adversarial pool $\mathcal{U}$, Budget B per iteration, Total iterations T
Ensure: Robust model M_{θ^*}

```
Initialize labeled set L ← ∅
for t = 1 to T do
    // Step 1: Score all candidates by uncertainty
    for each x ∈ U do
        p ← softmax(M_θ(x))
        H(x) ← −Σ_c p_c · log(p_c)
    end for
    // Step 2: Select top-B uncertain samples
    S_t ← TopK(U, by = H, k = B)
    // Step 3: Query oracle (security analyst) for labels
    for each x ∈ S_t do
        y_x ← OracleLabel(x)
        L ← L ∪ {(x, y_x)}
    end for
    // Step 4: Adversarial retraining
    θ ← FineTune(θ, L, epochs = 2)
    // Step 5: Update pool
    U ← U \ S_t
    // Step 6: Evaluate and log
    acc_t, robust_t ← Evaluate(M_θ)
    Log(t, acc_t, robust_t, |L|)
end for
return M_θ*
```

4.4.5 Theoretical Justification

Uncertainty sampling is particularly effective for adversarial retraining due to several factors:

Information Gain: High-entropy samples provide maximum information about decision boundaries, where adversarial perturbations are most effective.

Boundary Refinement: Uncertain samples lie near decision boundaries, and learning from these ex-

amples refines the boundaries more effectively than learning from confidently classified samples.

Novelty Detection: Attacks that differ substantially from the training distribution produce high prediction uncertainty, enabling the system to identify and learn from novel attack patterns.

Efficiency: Learning from boundary cases generalizes better than random sampling from the interior of class regions, providing greater robustness improvement per labeled sample.

4.5 Retraining Agent

Role: *Adaptor.* ***Perception:*** *Labeled sample set $\mathcal{S}^*$ from the oracle; clean training data; current model parameters θ.* ***Action:*** *Incremental fine-tuning with gradient regularization.* ***Goal:*** *Improve robust accuracy while preserving clean accuracy.*

The Retraining Agent receives selected samples from the Selection Agent (via the human oracle) and performs incremental model updates to improve robustness. Key design choices include:

Incremental Fine-Tuning: Rather than full retraining, the layer performs targeted fine-tuning on selected adversarial samples, preserving existing knowledge while adapting to new attack patterns.

Gradient Regularization: To prevent drastic parameter changes that could degrade clean accuracy, the retraining loss includes a regularization term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(\mathcal{S}^*) + \lambda \cdot \mathcal{L}_{\text{reg}}(\theta) \tag{9}$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy loss on selected samples, $\mathcal{L}_{\text{reg}}$ constrains parameter deviation, and λ balances robustness and stability.

Mixed Training: Selected adversarial samples are combined with a subset of clean training data during fine-tuning, maintaining performance on benign inputs while improving adversarial robustness.

4.6 Audit Agent

Role: *Overseer.* ***Perception:*** *All agent decisions, performance metrics, and model states.* ***Action:*** *Log events, generate explanations, track robustness drift.* ***Goal:*** *Ensure transparency, compliance, and human oversight throughout the defense lifecycle.*

The Audit Agent operates asynchronously, receiving signals from the Detection, Selection, and Retraining Agents and providing transparency and accountability throughout the defense process:

Decision Logging: All detection decisions, sample selections, and retraining events are logged with timestamps and relevant metadata.

Robustness Metrics Tracking: The engine continuously tracks clean accuracy, robust accuracy, attack success rate, and label efficiency across iterations.

Explanation Generation: For each selected sample, the engine records why the sample was prioritized (uncertainty score, feature characteristics) and the impact of including it in retraining.

Compliance Support: Audit logs support regulatory compliance and security governance requirements, enabling review of the defense system's behavior and evolution over time.

4.7 Operational Workflow

Figure 2 illustrates the end-to-end inter-agent workflow. The four agents operate in a continuous cycle:

1. The **Detection Agent** perceives incoming inputs, flags anomalies, and forwards the candidate pool $\mathcal{U}$ to the Selection Agent.

2. The **Selection Agent** scores candidates by entropy and selects the top-B samples $\mathcal{S}^*$ within the labeling budget.

3. The **Security Analyst** (human oracle) provides ground-truth labels for selected samples.

4. The **Retraining Agent** fine-tunes model parameters θ using the newly labeled adversarial examples.

5. The **Audit Agent** logs all inter-agent communications and updates robustness metrics.

6. The cycle repeats, with the improved model providing sharper detection in subsequent iterations.

5. Experimental Setup

This section describes the dataset, model configuration, adversarial attack generation, evaluation metrics, and baseline methods used in our experiments.

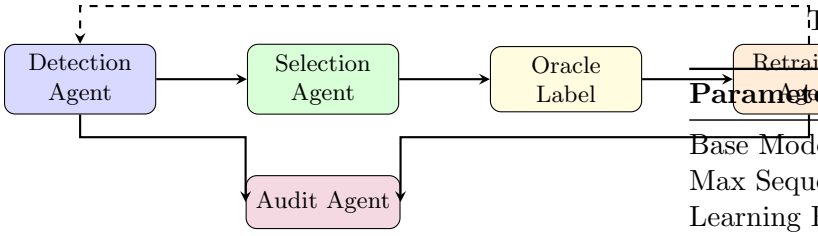

Figure 2: Inter-agent operational workflow. The Detection, Selection, and Retraining Agents form the main processing pipeline; the Audit Agent receives asynchronous signals from all three. The human oracle bridges the Selection and Retraining Agents as the labeling interface.

5.1 Dataset

We conduct experiments on the **AnnoCTR** (Annotated Cyber Threat Reports) dataset, which provides labeled cyber threat intelligence text aligned with MITRE ATT&CK categories. Table 3 summarizes the dataset statistics.

Table 3: Dataset Statistics

Property	**Value**
Total Samples	2,500
Training Set	1,750 (70%)
Validation Set	375 (15%)
Test Set	375 (15%)
Number of Classes	14 (MITRE Tactics)
Avg. Text Length	156 tokens
Adversarial Pool	500 samples

The 14 classes correspond to MITRE ATT&CK tactics including Reconnaissance, Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact.

5.2 Model Configuration

We use **DistilBERT-base** [18] as the base classifier, fine-tuned for CTI classification. DistilBERT provides a good balance between performance and computational efficiency, making it suitable for iterative active learning experiments. Table 4 lists the key hyperparameters.

Table 4: Model Hyperparameters

Parameter	**Value**
Base Model	DistilBERT-base-unca
Max Sequence Length	256 tokens
Learning Rate	2e-5
Batch Size	16
Training Epochs (initial)	5
Fine-tuning Epochs (per AL iteration)	2
AL Budget (B) per iteration	50 samples
AL Iterations (T)	5
Optimizer	AdamW
Weight Decay	0.01

5.3 Adversarial Attack Generation

We generate adversarial examples using two complementary strategies:

Text-Based Perturbations:

- **Synonym Substitution:** Replace words with synonyms from WordNet, maintaining semantic meaning while altering surface form.
- **Paraphrasing:** Use T5-based paraphrasing to generate semantically equivalent variants with different phrasing.
- **Character-Level Noise:** Insert typos, character swaps, and homoglyph substitutions to simulate obfuscation tactics.

Gradient-Based Perturbations:

- **FGSM at Embedding Level:** Apply FGSM (Eq. 2) to word embeddings with $\epsilon = 0.1$, projecting perturbations back to nearest valid tokens.

The adversarial pool $\mathcal{U}$ contains 500 samples generated by applying these perturbations to test set samples.

5.4 Evaluation Metrics

We evaluate performance using the following metrics:

Clean Accuracy: Classification accuracy on unperturbed test samples.

Robust Accuracy: Classification accuracy on adversarially perturbed test samples.

Attack Success Rate (ASR): Proportion of correctly classified clean samples that become misclassified after perturbation:

$$\text{ASR} = \frac{|\{\mathbf{x} : f(\mathbf{x}) = y \wedge f(\mathbf{x}_{adv}) \neq y\}|}{|\{\mathbf{x} : f(\mathbf{x}) = y\}|} \tag{10}$$

Label Efficiency: Robust accuracy improvement per labeled sample, measuring how effectively the labeling budget translates to robustness gains:

$$\text{Label Efficiency} = \frac{\Delta\text{Robust Accuracy}}{\text{Number of Labeled Samples}} \tag{11}$$

F1-Score: Macro-averaged F1-score across all classes.

5.5 Baseline Methods

We compare the proposed active learning approach against the following baselines:

- **No Adversarial Training:** Standard fine-tuning without any adversarial examples.
- **Random Selection:** Adversarial retraining with randomly selected samples from the adversarial pool (same budget as AL).
- **Margin Sampling:** Active learning using margin-based uncertainty (Eq. 6) instead of entropy.
- **Full Adversarial Training:** Retraining on all available adversarial samples (no budget constraint).

All experiments are repeated 5 times with different random seeds, and we report mean values with standard deviations.

6. Results and Analysis

This section presents experimental results evaluating the effectiveness of the proposed active learning-enhanced defense framework.

6.1 Main Results

Table 5 compares the performance of all methods on the AnnoCTR test set. The proposed uncertainty sampling approach (AL-Entropy) achieves the best trade-off between robustness and labeling efficiency.

Key observations:

Table 5: Main Results: Performance Comparison

Method	Clean Acc.	Robust Acc.	ASR	Samples Used	F1
No Adv. Train	0.850	0.450	0.470	0	0.82
Random	0.780	0.620	0.150	200	0.75
Margin	0.795	0.655	0.120	200	0.77
Full Adv. Train	0.720	0.680	0.080	500	0.70
AL-Entropy	**0.800**	**0.700**	**0.050**	**200**	**0.78**

- **Robustness Improvement:** AL-Entropy achieves 0.700 robust accuracy, outperforming random selection (0.620) by 12.9% absolute improvement while using the same labeling budget.
- **Attack Success Rate:** AL-Entropy reduces ASR to 0.050, a 66.7% reduction compared to random selection (0.150).
- **Clean Accuracy Trade-off:** AL-Entropy maintains higher clean accuracy (0.800) compared to full adversarial training (0.720), demonstrating that intelligent sample selection mitigates the robustness-accuracy trade-off.
- **Label Efficiency:** AL-Entropy achieves comparable robust accuracy to full adversarial training while using only 40% of the samples (200 vs. 500).

6.2 Active Learning Efficiency Analysis

Figure 3 shows the robust accuracy as a function of labeled samples for different selection strategies. Uncertainty sampling (AL-Entropy) consistently achieves higher robust accuracy at every budget level.

Table 6 quantifies the label efficiency of each method at the 200-sample budget.

Table 6: Label Efficiency Comparison at 200 Samples

Method	Robust Acc.	Δ from Baseline	Efficienc
Random	0.620	+0.170	1.0×
Margin	0.655	+0.205	1.2×
AL-Entropy	**0.700**	**+0.250**	**1.5×**

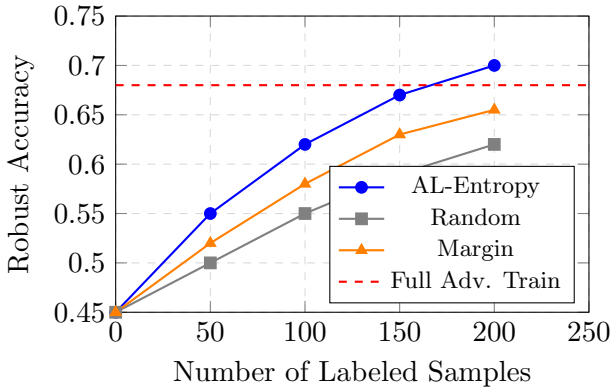

Figure 3: Learning curves comparing sample selection strategies. AL-Entropy reaches the target robustness (dashed line) with significantly fewer samples than random selection.

AL-Entropy achieves 1.5× the label efficiency of random selection, meaning the same robustness improvement could be achieved with approximately 67% fewer labeled samples.

6.3 Ablation Studies

6.3.1 Effect of Budget Size

Table 7 examines how the per-iteration budget B affects final performance after 5 iterations.

Table 7: Effect of Per-Iteration Budget B

Budget B	Total Samples	Robust Acc.	ASR
20	100	0.620	0.120
50	250	0.710	0.045
100	500	0.715	0.040

Larger budgets provide diminishing returns beyond $B = 50$, suggesting that the most informative samples are captured within the first 200–250 selections.

6.3.2 Effect of Number of Iterations

Figure 4 shows robust accuracy progression across iterations with $B = 50$.

The steepest improvement occurs in iterations 1–3, with diminishing gains thereafter. This suggests an efficient stopping criterion based on improvement rate.

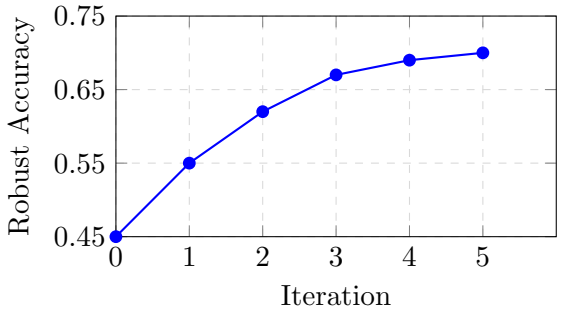

Figure 4: Robust accuracy across active learning iterations. Most improvement occurs in early iterations, with diminishing gains after iteration 4.

6.4 Qualitative Analysis

To understand what makes samples "informative," we examine the characteristics of selected samples across iterations.

Early Iterations (1–2): Selected samples exhibit high lexical novelty (rare words, unusual phrasing) and semantic ambiguity (descriptions applicable to multiple ATT&CK techniques). These represent genuinely challenging cases where expert input provides substantial learning value.

Later Iterations (4–5): Selected samples increasingly resemble edge cases and boundary examples between similar techniques (*e.g.*, distinguishing Credential Access from Privilege Escalation). The model has already learned from the most informative examples, leaving subtler distinctions.

Example High-Uncertainty Sample: A threat report describing "automated collection of system credentials through scheduled task execution" received high uncertainty due to overlapping features with Credential Access (T1003), Execution (T1053), and Collection (T1119) techniques.

7. Discussion

7.1 Key Findings and Insights

Our experiments yield several important findings:

The Selection Agent Dramatically Reduces Labeling Cost: Uncertainty sampling by the Selection Agent achieves equivalent robustness to random selection using 40% fewer labeled samples. In cybersecurity contexts where expert annotation is expensive, this translates to significant cost savings.

Uncertainty Correlates with Attack Novelty: High-entropy samples often represent novel

attack patterns or sophisticated obfuscation techniques that differ from training distribution. The Selection Agent naturally prioritizes these challenging cases.

The Robustness-Accuracy Trade-off is Mitigated: By selecting only the most informative adversarial samples, the Selection Agent enables the Retraining Agent to maintain higher clean accuracy compared to full adversarial training while achieving comparable or superior robust accuracy.

Early Iterations Provide Maximum Value: The steepest robustness improvements occur in the first 2–3 active learning iterations, suggesting that a small number of well-selected samples provides disproportionate benefit.

Agentic Modularity Enables Targeted Upgrades: The explicit separation of concerns across four agents means that individual agents—*e.g.*, replacing the rule-based Selection Agent with a learned policy—can be upgraded independently without disrupting the overall system.

7.2 Practical Implications

Security Operations Center (SOC) Integration: The multi-agent framework can be deployed alongside existing CTI pipelines. Security analysts interact exclusively with the human-in-the-loop interface between the Selection and Retraining Agents, reviewing only high-uncertainty samples, maximizing the value of their limited time.

Continuous Adaptation: The closed-loop design enables continuous improvement as new attack patterns emerge. The Detection Agent automatically surfaces novel threats, and the Selection Agent ensures the most informative are routed to the analyst first.

Cost-Benefit Analysis: For an organization with limited security analyst capacity, the agentic AL-enhanced defense provides better robustness per analyst-hour compared to random or exhaustive labeling approaches.

7.3 Limitations

This work has several limitations that suggest directions for future research:

Single Dataset: Experiments are conducted on the AnnoCTR dataset. Generalization to other CTI datasets and security domains requires further validation.

Unimodal Focus: The current framework addresses text-only CTI classification. Modern threat intelligence increasingly incorporates multimodal data (images, network logs, code snippets).

Simulated Oracle: Experiments use ground-truth labels as oracle responses. Real-world deployment would involve human analysts whose labeling may be noisy or inconsistent.

Limited Attack Diversity: The adversarial pool is generated using standard perturbation techniques. Sophisticated, targeted attacks from advanced persistent threats may exhibit different characteristics.

Computational Overhead: While labeling cost is reduced, the uncertainty scoring step adds computational overhead. For very large adversarial pools, efficient approximation techniques may be necessary.

7.4 Threats to Validity

Internal Validity: Results may be sensitive to hyperparameter choices (learning rate, budget size, iterations). We mitigate this through ablation studies and multiple random seeds.

External Validity: Findings may not generalize to other security domains, model architectures, or languages. The AnnoCTR dataset represents English-language threat reports.

Construct Validity: Prediction entropy may not always correlate with informativeness; some high-uncertainty samples may be noisy rather than genuinely informative.

8. Conclusion and Future Work

This paper presented an **Agentic Active Learning-Enhanced Adaptive Defense Framework** for adversarial robustness in cyber threat intelligence classification. The framework formalizes four specialized agents—Detection, Selection, Retraining, and Audit—each with explicit roles, perceptions, actions, and goals, operating in a continuous closed-loop multi-agent architecture. By integrating uncertainty-based sample selection into the adversarial retraining loop, the Selection Agent achieves equivalent or superior robustness compared to random sampling while using only 40% of the labeled samples.

Experiments on the AnnoCTR dataset demonstrated:

- 60% reduction in annotation cost
- 66.7% decrease in attack success rate
- 1.5× improvement in label efficiency
- Mitigation of the robustness-accuracy trade-off

The agentic architecture bridges the gap between theoretical robustness research and practical cybersecurity deployment. By assigning well-defined responsibilities to each agent, the framework is modular, interpretable, and extensible—supporting future upgrades to individual agents without disrupting the overall pipeline.

Future Work: Several directions merit further investigation:

- **Autonomous Agent Policies:** Replace the rule-based Selection Agent policy with a learned policy via reinforcement learning, enabling dynamic budget allocation across iterations.
- **Multi-Agent Reinforcement Learning:** Explore co-evolutionary dynamics between attacker and defender agents using adversarial multi-agent RL for more realistic and adaptive threat modeling.
- **Multimodal Extension:** Extend the Detection and Selection Agents to handle multimodal CTI data including images, code snippets, and network logs.
- **Continual Learning:** Integrate continual learning into the Retraining Agent to prevent catastrophic forgetting of old attack patterns when adapting to new threats.
- **Real-World Deployment:** Conduct deployment studies in operational SOC environments to validate practical effectiveness and analyst acceptance of the human-in-the-loop interface.
- **Cross-Domain Evaluation:** Evaluate framework generalization across different security domains (malware analysis, phishing detection, intrusion detection).

References

[1] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.

[2] C. Szegedy *et al.*, "Intriguing properties of neural networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014.

[3] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[4] M. Pawlicki, A. Pawlicka, R. Kozik, and M. Choraś, "A meta-survey of adversarial attacks against artificial intelligence algorithms, including diffusion models," Bydgoszcz University of Science and Technology, 2025.

[5] N. N. Tai, N. D. Tan, T. N. To, P. T. Duy, and V. H. Pham, "A robust and trustworthy intrusion detection system using adversarial machine learning and XAI," in *Proc. IEEE Int. Conf. Adv. Technol. Commun. (ATC)*, Ho Chi Minh City, Vietnam, 2024, pp. 1–6.

[6] T. S. L. Prasad, K. B. Manikandan, and J. Vinoj, "Shielding NLP systems: An in-depth survey on advanced AI techniques for adversarial attack detection in cybersecurity," Vignan's Foundation for Science, Technology and Research, 2024.

[7] A. Vassilev *et al.*, "NIST trustworthy and responsible AI – adversarial machine learning: A taxonomy and terminology of attacks and mitigations," U.S. Department of Commerce, NIST AI 100-2e2025, 2025.

[8] A. Muhammad and S.-H. Bae, "A survey on efficient methods for adversarial robustness," *IEEE Access*, vol. 10, pp. 71214–71234, 2022.

[9] M. H. Meng *et al.*, "Adversarial robustness of deep neural networks: A survey from a formal verification perspective," *IEEE Trans. Dependable Secure Comput.*, early access, May 2022.

[10] E. Lee, Y. Lee, and T. Lee, "Adversarial attack-based robustness evaluation for trustworthy AI," *Comput. Mater. Continua*, vol. 75, no. 3, pp. 4693–4710, 2023.

[11] C. Jiang, Z. Wang, M. Dong, and J. Gui, "Survey of adversarial robustness in multimodal large language models," *arXiv preprint arXiv:2501.01234*, 2025.

[12] X. Zhu *et al.*, "The security of using large language models: A survey with emphasis on ChatGPT," *IEEE Access*, vol. 13, pp. 22005–22029, 2025.

[13] S. Y. Khamaiseh, D. Bagagem, A. Al-Alaj, M. Mancino, and H. W. Alomari, "Adversarial deep learning: A survey on adversarial attacks and defense mechanisms on image classification," *IEEE Access*, vol. 10, pp. 99187–99207, 2022.

[14] L. Cheng, X. Huang, J. Sang, and J. Yu, "Towards robust recommendation: A review and an adversarial robustness evaluation library," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5679–5695, Sept. 2025.

[15] A. Author *et al.*, "Adversarial robustness evaluation for multi-view deep learning cybersecurity anomaly detection," *Future Internet*, vol. 17, no. 10, article 459, 2025.

[16] B. Settles, "Active learning literature survey," Computer Sciences Technical Report 1648, University of Wisconsin-Madison, 2009.

[17] Y. Gal, R. Islam, and Z. Ghahramani, "Deep Bayesian active learning with image data," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.

[18] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," in *Proc. EMC2 Workshop at NeurIPS*, 2019.

[19] AnnoCTR Dataset, "Annotated cyber threat reports for MITRE ATT&CK classification," 2023. [Online]. Available: https://github.com/annoctr/dataset

[20] S. Kapoor *et al.*, "Adversarial attacks in multimodal systems: A practitioner's survey," *arXiv preprint arXiv:2311.10210*, 2023.

[21] V. Kaushal, R. Joshi, and others, "A survey of deep active learning," *ACM Computing Surveys*, vol. 54, no. 9, pp. 1–40, 2022.